



3 Payload Demonstration Flight



Figure 15: Assembled payload prior to launch

3.1 Payload Summary

The payload mission was partially successful during this launch. The rocket landed in a tree, so we were unable to collect a soil sample. As a result, we were not able to test our soil conductivity sensor in mission conditions. However, we were able to verify that the excavator motor did turn on after landing, though it did not turn off until much later than we intended. We had to remove the springs prior to flight as they collided with the flaps, preventing them from fully closing. So, the flaps were freely-pivoting. We added extra nuts to the lower half of the threaded rods to add extra mass, bringing our mass closer to the mass flown during VDF. The total as-flown weight came out to be 3.40 lbs, including the quick links at the top.



Figure 16: Hinge detail before launch. Note the aluminum flap tab



3.2 Payload Mission Sequence

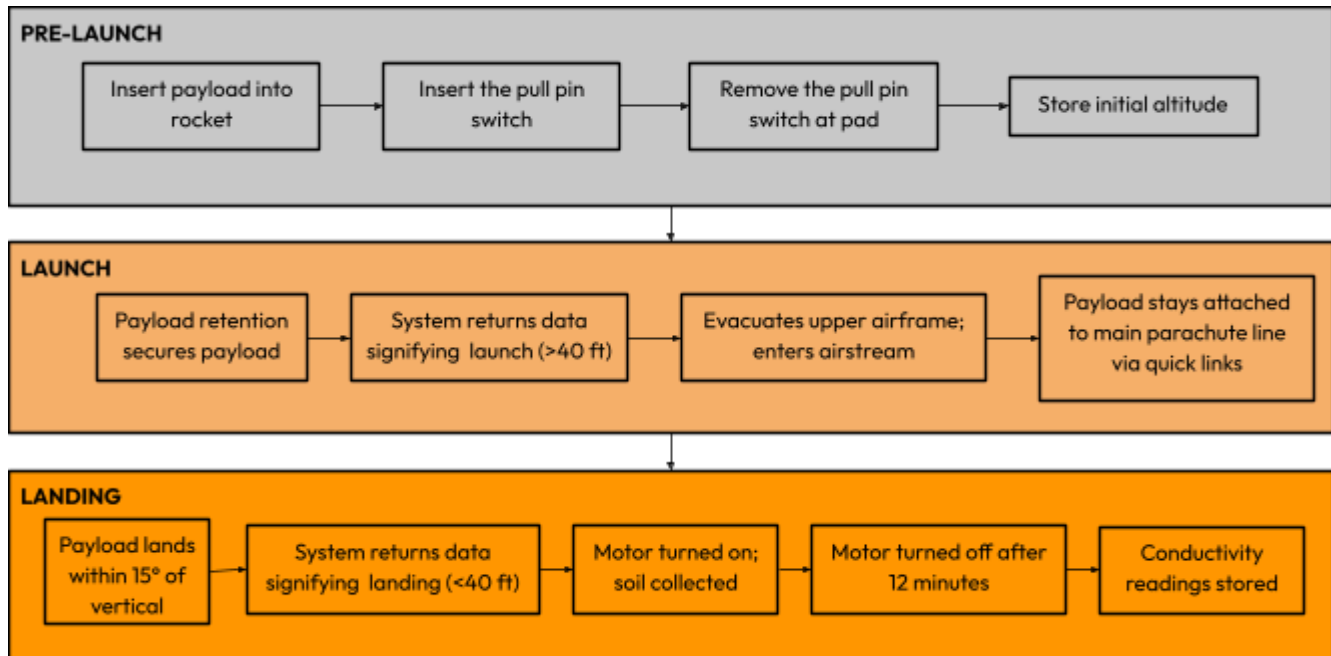


Figure 17: Payload mission sequence diagram

3.3 Payload Retention

During flight, the payload capsule sits on a pair of blocks that prevent it from falling downwards. Another pair of blocks positioned forward of the payload prevent the payload from sliding forwards into the nose cone. This system is the same as in the VDF flight and is detailed in section 4.1.1. The payload itself is positioned upside down during flight, with the STEMnaut quarters facing to the aft end of the rocket. The eye bolt extending out of the STEMnaut quarters is secured to the main parachute line by quick links. During the PDF, two quick links were used to increase the payload's overall mass and thus improve the rocket's stability.

3.4 Altimeter Data



Altitude (m) vs. Time After Launch (s)

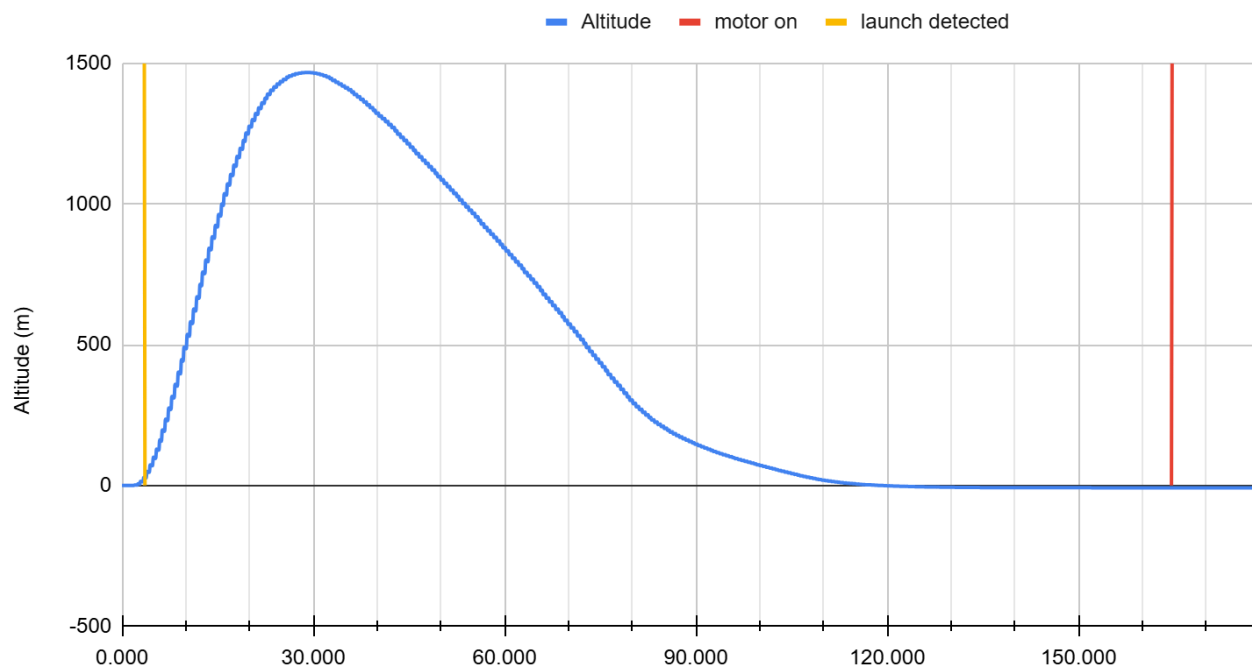


Figure 18: Payload Altimeter Data

The yellow bar indicates when the altimeter detected launch, and the red bar indicates when the motor is turned on. The motor is turned on about thirty seconds after launch because the motor controls requires the payload capsule to be stable before turning on the motor.

3.5 System-Level Review

3.5.1 Nominally-Functioning Systems

Most of the system functioned as intended. The data logging was successful since the altitude data, time stamp, and the state of the motor are stored and visible in the SD card. Additionally, it is applicable to recover all the data after flight.

The motor is verified to be on because every time it is triggered, the corresponding state is logged onto the SD card. The motor is visually on around 160 seconds after the launch, according to Figure 18.

The launch and landing are detected successfully, which are also solely based on the height/altitude detection. The motor is evident to prove the functioning of the launch and landing detection process.



Before and during the launch, the pull pin was working properly but after the launch we do have problems with resetting it, see section 3.5.2.

The battery is enough for the entire flight duration and as it lasted more than 2 hours. There are data and time stamps stored within the SD card to know the amount of time that the battery is able to support the system.

The electronics retention systems functioned as intended because they remained secure during the entire flight. Wires were bundled together with electrical tape, and the electronics boards were contained inside a 3D printed box. This box was secured to the electronics bay bulkhead with 4 bolts, and the lid secured to the box with another 4 bolts. The lid had holes drilled for wires to exit. For out-of-this-world adhesion, the 12V battery was first secured to the electronics bay bulkhead surface with Alien Tape, which is a double-sided all-surface tape. The 9V battery was attached to the side of the 12V battery with Alien Tape and duct tape as well. Then, both batteries were tied down with 4 zip ties that went through holes drilled into the bulkhead.

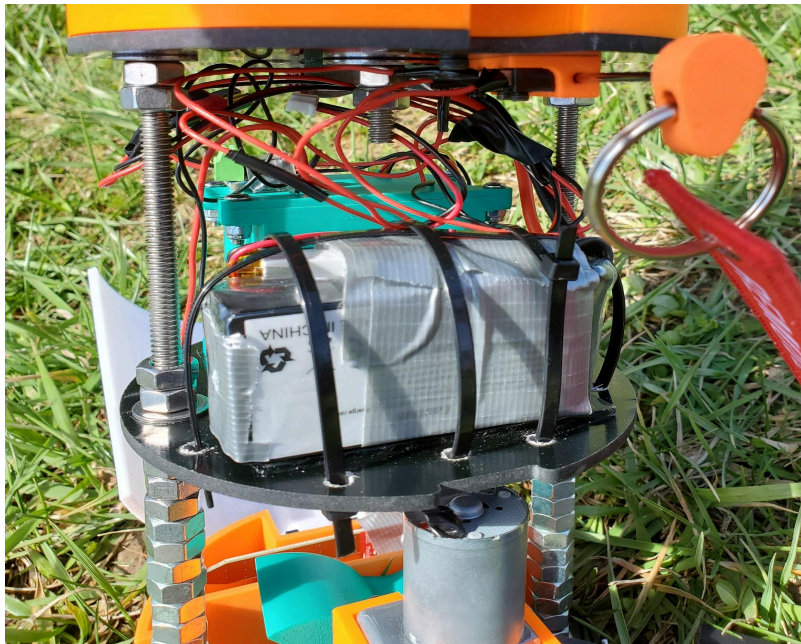


Figure 19: Battery tie-downs



Figure 20: Lower level detail. Left: Motor; Center: Excavator; Right: Soil collection chamber with conductivity sensor

3.5.2 Off-Nominal Systems

Our pull pin was not easily resettable, as putting the pin back in did not immediately shut off power. The pin needed to be pressed upwards to actually engage the switch. This is likely something stuck internally, and will require some disassembly to diagnose and fix.

The motor ran for two hours straight due to a bug in the code. Instead of checking altitude and launch condition, the code just checked for altitude leading to the motor running while on the ground pre launch. The motor turned off during launch and didn't impact the flight of the rocket, then it turned back on after the rocket landed. This is a simple fix in the code, checking the boolean value launched is true as part of turning on the motor.

Our data on the SD card was difficult to recover due to a bug in the code. Instead of writing to a new file each time the system was powering up, it was overwriting old data. This is also a simple fix in the code, making it write to a new file each time as opposed to overwriting old data. We were able to recover the deleted data on the SD card, and we fixed the bug to avoid this problem in the future.

We were unable to properly test our soil collection mechanism due to landing in a tree. There is no soil in the air, so the excavator was unable to collect any soil. Ground testing of the excavator will be conducted to ensure that the excavator is functioning properly. We were unable to obtain a conductivity reading due to the lack of soil in the soil collection chamber.

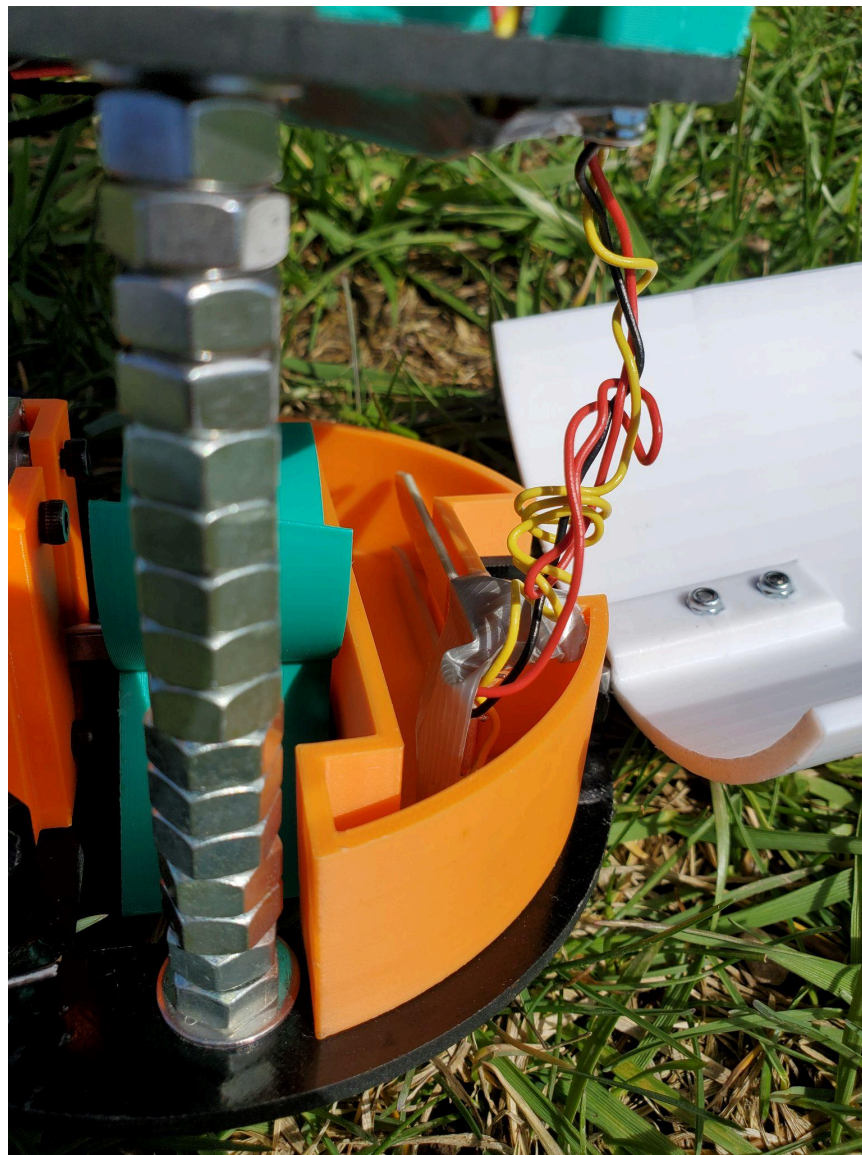


Figure 21: Soil sensor detail

3.5.3 Damaged Components

The only damaged components were two zip ties that secured the battery, but it was still held in place by the double-sided Alien Tape. There were a total of four zip ties, so two remained upon landing. We cannot determine when these zip ties broke, as the payload fell somewhat roughly out of the tree during recovery.

The pull pin switch may have been damaged sometime between assembly in the shop and being loaded into the rocket. We noticed that the pin had to be manually adjusted while the pin was fully inside the switch in order to engage the contacts. This was not an issue when we tested the electronics in the shop prior to launch. We will use one of our spare back-up switches for the next launch.

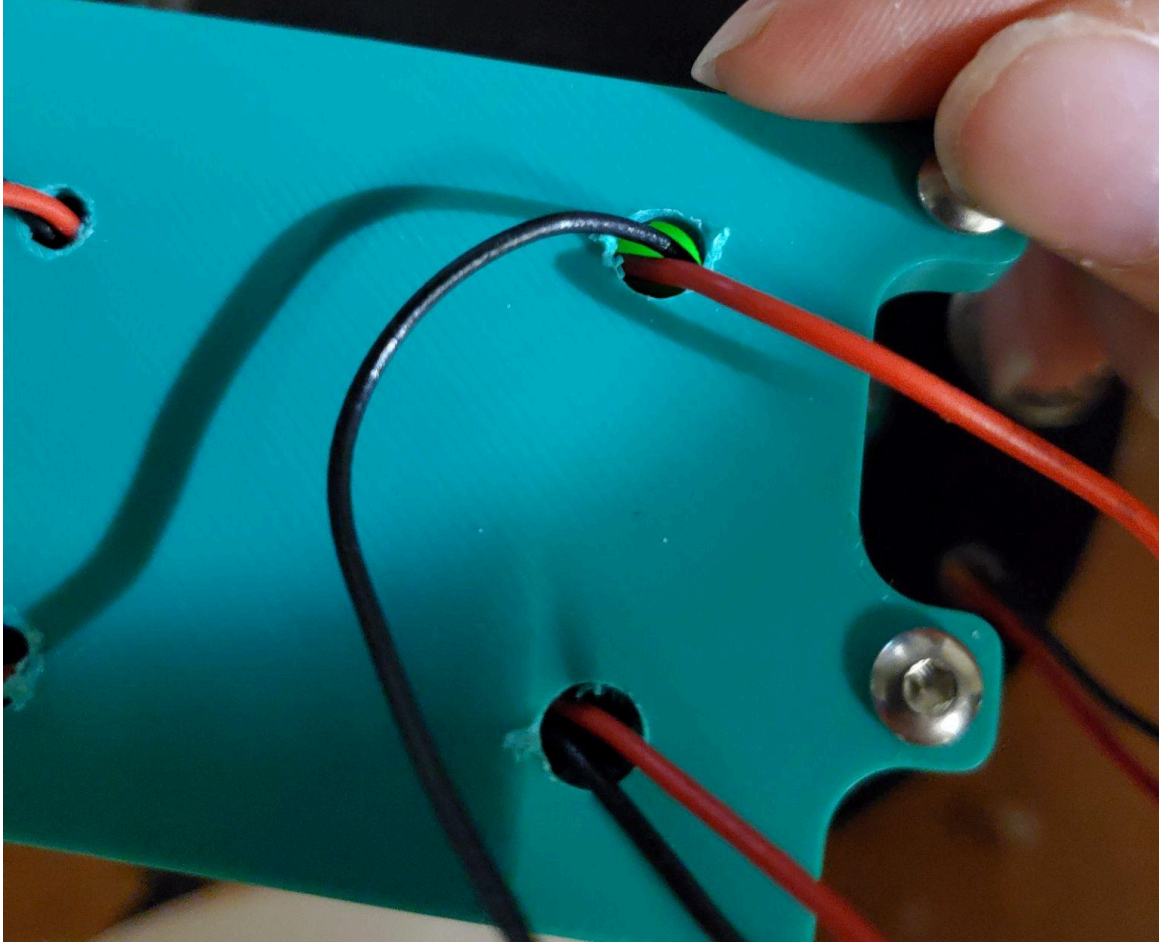


Figure 22: Successful electronics test in the shop, indicated by the green light (pictured) and flashing white lights (not pictured)

3.6 Payload Lessons Learned

One of the major mechanical issues were the hinges on the payload flaps. The springs were providing insufficient force when the flaps were at the lowest position below the horizontal. Because of this, the payload would be unable to upright if it lands on its side.

The solution to this would be a hinge redesign on the payload. We will use modified [surface-mount light duty spring hinges](#) to provide sufficient force for the flaps at the lowest position. From ejection testing, the hinge system must be robust otherwise it breaks as it hits the ground. One of the hinge leaves will be mounted to fiberglass posts on the payload base and provide a robust structure for flap attachment. The other leaf of the hinge will be attached to the flap. To ensure resistance against vibrational loads, M3 locknuts will be used as the attachment from flap to hinge.

A major software issue was the SD card writing control. The software would replace the log file every time it restarts. If during recovery power gets flashed, like during this current run